Paulo Quirino

Lead Frontend Engineer & Systems Architect

Specialist in React, TypeScript, Component Systems & High-Performance Web Engines

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PROFILE

Bridges low-level mechanical sympathy (C99, WASM, POSIX event loops) with modern reactive UI ergonomics (React, TypeScript, state orchestration). 8+ years delivering high-scale web platforms, real-time robotics telemetry interfaces, and datatype-driven component systems with deterministic performance and minimal operational overhead.

EXPERIENCE

Front-end & Architecture Specialist
11/2025 - 08/2026

Loomtech - Lisbon

Architected frontend and data model for a block-based document editor engine supporting drag-and-drop layouts, dynamic formatting, and PDF export. Engineered datatype-driven UI framework automating component reuse across 4 core entities.

Stack: React, TypeScript, Valtio, React-Query, Zod, TailwindCSS, Vite

Lead Frontend Developer
11/2022 - 02/2025

Mov.ai - Lisbon

Primary frontend engineer for autonomous robotics solution. Built 3D interactive scenes in BabylonJS, a visual node flow behavior editor, and multi-type code editor for robotics coordination. Developed real-time telemetry tables and centralized log controller.

Stack: React 18, TypeScript, BabylonJS, Material-UI 6, Remix, Vite, esbuild

Front-end Developer / Tech Lead
12/2021 - 07/2022

PrimeIT - Lisbon

Tech lead and frontend engineer architecting micro-frontends for enterprise GDPR compliance platform. Pioneered datatype-driven UI architecture and pure-stack design principles, formalizing data models.

Stack: React, Micro-Frontends, React-Router, TypeScript, SCRUM

Full-stack Developer
10/2020 - 11/2021

UNPARALLEL - Lisbon

Delivered full-stack applications in Next.js. Integrated IoT telemetry with a custom C++ Arduino library for acoustic sensors and authored JavaScript utility libraries.

Stack: Next.js, React, TypeScript, C++, WordPress, Node.js

Full-stack Developer
11/2019 - 05/2020

Mobinteg - Lisbon

Architected an interoperable runtime launcher executing seamlessly across Angular and Mithril. Engineered an early utility-first CSS compilation engine predating modern Tailwind paradigms.

Stack: Angular, Mithril, TypeScript, Webpack, Babel, PHP, MySQL

Web & Mobile Developer
08/2017 - 08/2018

Exictos - Funchal

Maintained core banking employee portals and engineered cross-platform mobile clients with React Native and Ionic. Authored C tool for deduplicating localization assets.

Stack: React Native, Ionic, JavaScript, C, Spring Boot

CORE SKILLS

Frontend Architecture & UI

[React] [TypeScript] [Next.js] [Micro-Frontends]
[React-Query] [Zod] [Valtio] [TailwindCSS]
[Material-UI] [Redux] [BabylonJS (3D)]
[Component Reuse] [Vite]

Systems, C & Architecture

[C (C99/C11)] [C++] [Systems Architecture]
[POSIX/Unix] [Isomorphic WASM] [In-Memory DBs]
[Full-Text Search] [Data Modelling]

Backend & Databases

[Node.js] [PostgreSQL] [Redis] [MongoDB]
[BerkeleyDB] [MySQL] [PHP]

AI, Tooling & DevOps

[Python] [Vulkan/Local LLMs] [AI Workflows]
[Linux & DevOps] [Docker] [Github CI]

SYSTEMS ARCHITECTURE & PROJECTS

axil HTTP/WS Server

Event-driven HTTP(S) and WebSocket daemon in C99. Non-blocking I/O loop, zero dependencies.

libxylem Dynamic Module System

Hook-based plugin kernel with shared library DAG loading, lifecycle hooks, and ABI boundaries.

EDUCATION

Computer Science & Systems Engineering
University of Madeira

LANGUAGES

Portuguese	Native / C2
English	Fluent / C1/C2